



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

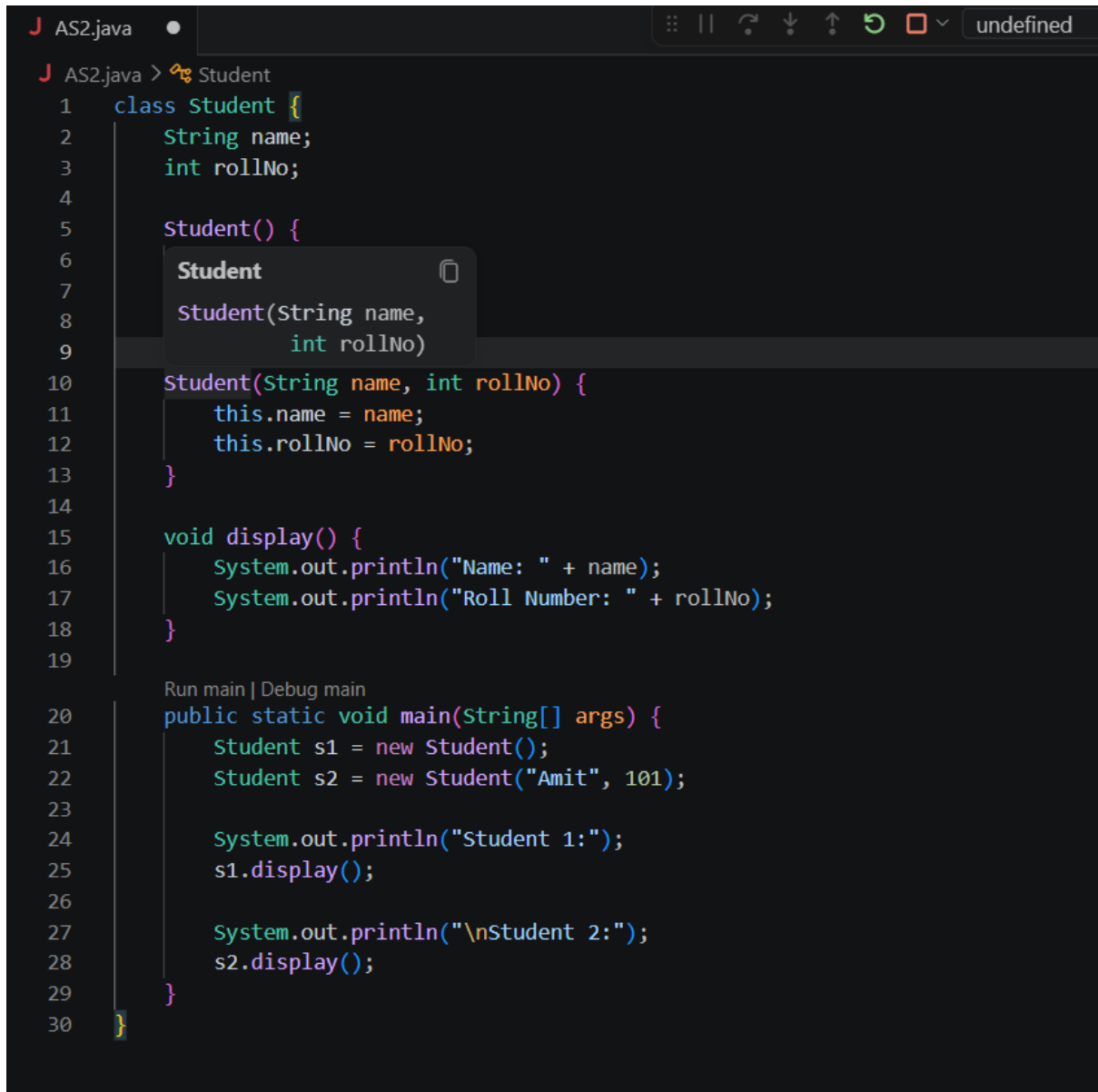
Outcome Conclusion

The assignment successfully demonstrated the use of default, parameterized, and copy constructors in Java. Constructors provide an efficient way to initialize objects with required values, while copy constructors help create duplicate objects without rewriting data. These

concepts are fundamental to object-oriented programming and improve code organization and maintainability.

Exercise

1. Create a **Student** class using default and parameterized constructors to initialize student name and roll number.



```
J AS2.java > Student
1  class Student {
2      String name;
3      int rollNo;
4
5      Student() {
6          Student
7          Student(String name,
8              int rollNo)
9
10     Student(String name, int rollNo) {
11         this.name = name;
12         this.rollNo = rollNo;
13     }
14
15     void display() {
16         System.out.println("Name: " + name);
17         System.out.println("Roll Number: " + rollNo);
18     }
19
20     Run main | Debug main
21     public static void main(String[] args) {
22         Student s1 = new Student();
23         Student s2 = new Student("Amit", 101);
24
25         System.out.println("Student 1:");
26         s1.display();
27
28         System.out.println("\nStudent 2:");
29         s2.display();
30     }
```

OUTPUT-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Filter (e.g. text, lexclude, \esca...)  Q undefined  X  X  X

Student 1:
Name: Unknown
Roll Number: 0

Student 2:
Name: Amit
Roll Number: 101
```

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

```
J AS2.java  J AS2EX2.java X
J AS2EX2.java > Mobile
1  class Mobile {
2      String brand;
3      String model;
4      double price;
5
6      Mobile() {
7          brand = "Unknown";
8          model = "Unknown";
9          price = 0.0;
10     }
11
12     Mobile(String brand, String model, double price) {
13         this.brand = brand;
14         this.model = model;
15         this.price = price;
16     }
17
18     Mobile(Mobile m) {
19         this.brand = m.brand;
20         this.model = m.model;
21         this.price = m.price;
22     }
23
24     void display() {
25         System.out.println("Brand : " + brand);
26         System.out.println("Model : " + model);
27         System.out.println("Price : " + price);
28     }
29
```

```
Run main | Debug main
30 public static void main(String[] args) {
31     Mobile m1 = new Mobile(); // Default constructor
32
33     Mobile m2 = new Mobile("Samsung", "Galaxy S24", 75000); // Parameterized constructor
34
35     Mobile m3 = new Mobile(m2); // Copy constructor
36
37     System.out.println("Mobile 1 Details:");
38     m1.display();
39
40     System.out.println("\nMobile 2 Details:");
41     m2.display();
42
43     System.out.println("\nMobile 3 Details (Duplicate Record):");
44     m3.display();
45 }
46 }
```

OUTPUT-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Mobile 1 Details:
Brand : Unknown
Model : Unknown
Price : 0.0

Mobile 2 Details:
Brand : Samsung
Model : Galaxy S24
Price : 75000.0

Mobile 3 Details (Duplicate Record):
Brand : Samsung
Model : Galaxy S24
Price : 75000.0
```